TA Session – Week 4

Expected Value, Random Variables and Probability Distributions

Q1. Ranking of Men and Women

Five men and five women are ranked according to their scores on an examination. Assume that no two scores are alike and all 10! possible rankings are equally likely. Let X denote the highest ranking achieved by a woman. For example, $X = 2$ if the top-ranked person is male and the second-ranked person is female. Compute $E[X]$.

Solution using $E[X] = \sum_x xP(X = x)$

X is the smallest position occupied by a woman. Therefore, X can take values $1, 2, \ldots, 6$.

$$P(X = 1) = \frac{5}{10} = \frac{1}{2}$$

$$P(X = 2) = \frac{5}{10}\frac{5}{9} = \frac{5}{18}$$

$$P(X = 3) = \frac{5}{10}\frac{4}{9}\frac{5}{8} = \frac{5}{36}$$

$$P(X = 4) = \frac{5}{10}\frac{4}{9}\frac{3}{8}\frac{5}{7} = \frac{5}{84}$$

$$P(X = 5) = \frac{5}{10}\frac{4}{9}\frac{3}{8}\frac{2}{7}\frac{5}{6} = \frac{1}{42}$$

$$P(X = 6) = \frac{5}{10}\frac{4}{9}\frac{3}{8}\frac{2}{7}\frac{1}{6} = \frac{1}{252}$$

Hence,

$$E[X] = 1\left(\frac{1}{2}\right) + 2\left(\frac{5}{18}\right) + 3\left(\frac{5}{36}\right) + 4\left(\frac{5}{84}\right) + 5\left(\frac{1}{42}\right) + 6\left(\frac{1}{252}\right)$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$E[X] = \frac{11}{6} \approx 1.833$$

$$\boxed{E[X] = \frac{11}{6} \approx 1.833}$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q2. Difference Between Heads and Tails

Define X as the difference between the number of heads and tails in three tosses of a fair coin. Find $E[X]$.

Solution using $E[X] = \sum_x xP(X = x)$

For three tosses, if H is the number of heads, then

$$T = 3 - H$$

and

$$X = H - T = H - (3 - H) = 2H - 3.$$

The possible values are:

H	X	$P(X)$
0	-3	$\frac{1}{8}$
1	-1	$\frac{3}{8}$
2	1	$\frac{3}{8}$
3	3	$\frac{1}{8}$

Therefore,

$$E[X] = (-3)\left(\frac{1}{8}\right) + (-1)\left(\frac{3}{8}\right) + (1)\left(\frac{3}{8}\right) + (3)\left(\frac{1}{8}\right)$$

$$E[X] = 0$$

$$\boxed{E[X] = 0}$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q3. Weather Forecaster Scoring

A weather forecaster announces a rain probability p. The score is

$$1 - (1 - p)^2$$

if rain occurs, and

$$1 - p^2$$

otherwise. If the true probability of rain is p^*, find the value of p that maximizes the expected score.

Solution

The true probabilities are

$$P(\text{rain}) = p^*$$

and

$$P(\text{no rain}) = 1 - p^*.$$

Therefore,

$$E[S] = p^* \left[1 - (1 - p)^2\right] + (1 - p^*)(1 - p^2).$$

Since

$$1 - (1 - p)^2 = 2p - p^2,$$

we have

$$E[S] = p^*(2p - p^2) + (1 - p^*)(1 - p^2).$$

Differentiating with respect to p,

$$\frac{dE[S]}{dp} = 2p^* - 2p.$$

Setting this equal to zero,

$$2p^* - 2p = 0$$

gives

$$p = p^*.$$

Also,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\frac{d^2E[S]}{dp^2} = -2 < 0.$$

Hence, this stationary point gives a maximum.

$$\boxed{p = p^*}$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q4. Insurance Policy

An insurance company writes a policy stating that an amount A must be paid if some event E occurs within a year. If the company estimates that E will occur within the year with probability p, what should it charge the customer so that its expected profit is 10% of A?

Solution using $E[X] = \sum_x xP(X = x)$

Let C be the premium and let X be the company's profit.

If E occurs, with probability p,

$$X = C - A.$$

If E does not occur, with probability $1 - p$,

$$X = C.$$

Therefore,

$$E[X] = (C - A)p + C(1 - p).$$

Simplifying,

$$E[X] = C - Ap.$$

The required expected profit is

$$0.10A.$$

Therefore,

$$C - Ap = 0.10A.$$

Hence,

$$C = Ap + 0.10A$$

and therefore

$$\boxed{C = (p + 0.10)A}.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q5. Bernoulli Random Variable

A transaction is fraudulent with probability $p = 0.05$. Let

$$X \sim \text{Ber}(0.05).$$

Find $E[X]$.

Solution

A Bernoulli random variable takes the values

$$X = \begin{cases} 1, & \text{with probability } 0.05, \\ 0, & \text{with probability } 0.95. \end{cases}$$

Thus,

$$E[X] = 1(0.05) + 0(0.95) = 0.05.$$

$$\boxed{E[X] = 0.05}$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q6. Expected Number of Fraudulent Transactions

You review 100 independent transactions, each having a 5% chance of being fraudulent. Let

$$N \sim \text{Bin}(100, 0.05).$$

Find the expected number of fraudulent transactions, $E[N]$.

Solution

For a binomial random variable,

$$E[N] = np.$$

Therefore,

$$E[N] = 100(0.05) = 5.$$

$$\boxed{E[N] = 5}$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q7. Uniform Random Variable

Let

$$X \sim \text{Unif}(0, 10).$$

Find the expected value $E[X]$.

Solution

For a continuous uniform random variable

$$X \sim \text{Unif}(a, b),$$

the expected value is

$$E[X] = \frac{a+b}{2}.$$

Therefore,

$$E[X] = \frac{0+10}{2} = 5.$$

$$\boxed{E[X] = 5}$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q8. Students and Buses

A total of four buses carrying 148 students from the same school arrive at a football stadium. The buses carry, respectively,

$$40, \quad 33, \quad 25, \quad 50$$

students.

One student is randomly selected. Let X denote the number of students on the bus carrying the randomly selected student.

One of the four bus drivers is also randomly selected. Let Y denote the number of students on her bus.

(a) Which of $E[X]$ or $E[Y]$ is larger? Why?

(b) Compute $E[X]$ and $E[Y]$.

Solution

A randomly chosen student is more likely to belong to a larger bus. Therefore,

$$E[X] > E[Y].$$

For X,

$$P(X = 40) = \frac{40}{148},$$

$$P(X = 33) = \frac{33}{148},$$

$$P(X = 25) = \frac{25}{148},$$

and

$$P(X = 50) = \frac{50}{148}.$$

Hence,

$$E[X] = 40\left(\frac{40}{148}\right) + 33\left(\frac{33}{148}\right) + 25\left(\frac{25}{148}\right) + 50\left(\frac{50}{148}\right)$$

$$E[X] = \frac{5814}{148} \approx 39.28.$$

For Y, each driver is equally likely to be selected. Thus,

$$E[Y] = \frac{40 + 33 + 25 + 50}{4} = 37.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Therefore,

$$\boxed{E[X] \approx 39.28, \qquad E[Y] = 37}$$

and

$$\boxed{E[X] > E[Y]}.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q9. Expected Number of Games

Suppose two teams play a series of games that ends when one of them has won two games. Suppose that each game is independently won by team A with probability p. Find the expected number of games played. Also show that this expected number is maximized when $p = 0.5$.

Solution

The series lasts either 2 or 3 games.

The probability that the series ends after two games is

$$P(N = 2) = p^2 + (1 - p)^2.$$

The probability that it lasts three games is

$$P(N = 3) = 2p(1 - p).$$

Using

$$E[N] = \sum_n nP(N = n),$$

we obtain

$$E[N] = 2\left[p^2 + (1 - p)^2\right] + 3\left[2p(1 - p)\right].$$

Simplifying,

$$E[N] = 2 + 2p(1 - p).$$

Differentiate with respect to p:

$$\frac{dE[N]}{dp} = 2 - 4p.$$

Setting this equal to zero,

$$2 - 4p = 0$$

gives

$$p = \frac{1}{2}.$$

The second derivative is

$$\frac{d^2E[N]}{dp^2} = -4 < 0,$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

so this is a maximum.
At $p = 0.5$,

$$E[N] = 2 + 2(0.5)(0.5) = 2.5.$$

Therefore,

$$\boxed{E[N] = 2 + 2p(1-p)}$$

and

$$\boxed{E[N]_{\max} = 2.5 \text{ at } p = 0.5}.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q10. Finding Parameters of a Probability Density Function

A continuous random variable has density

$$f(x) = a + bx^2, \qquad 0 \le x \le 1,$$

and zero otherwise. Given that

$$E[X] = \frac{3}{5},$$

determine a and b.

Solution

Since $f(x)$ is a probability density function,

$$\int_0^1 (a + bx^2)\, dx = 1.$$

Therefore,

$$a + \frac{b}{3} = 1,$$

so

$$a = 1 - \frac{b}{3}.$$

Using the expected value,

$$E[X] = \int_0^1 x f(x)\, dx = \frac{3}{5}.$$

Thus,

$$\int_0^1 x(a + bx^2)\, dx = \frac{3}{5},$$

which gives

$$\frac{a}{2} + \frac{b}{4} = \frac{3}{5}.$$

Substituting

$$a = 1 - \frac{b}{3}$$

gives

$$b = \frac{6}{5}.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Therefore,

$$a = 1 - \frac{1}{3}\left(\frac{6}{5}\right) = \frac{3}{5}.$$

Hence,

$$\boxed{a = \frac{3}{5}, \qquad b = \frac{6}{5}}.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q11. Properties of Expectation

If

$$E[X] = 2$$

and

$$E[X^2] = 8,$$

calculate:

(a) $E[(2+4X)^2]$,

(b) $E[X^2 + (X+1)^2]$.

Solution

(a)

First expand:

$$(2+4X)^2 = 4 + 16X + 16X^2.$$

Therefore,

$$\begin{aligned} E[(2+4X)^2] &= 4 + 16E[X] + 16E[X^2] \\ &= 4 + 16(2) + 16(8) \\ &= 164. \end{aligned}$$

Thus,

$$\boxed{E[(2+4X)^2] = 164}.$$

(b)

First expand:

$$X^2 + (X+1)^2 = X^2 + X^2 + 2X + 1 = 2X^2 + 2X + 1.$$

Therefore,

$$\begin{aligned} E[X^2 + (X+1)^2] &= 2E[X^2] + 2E[X] + 1 \\ &= 2(8) + 2(2) + 1 \\ &= 21. \end{aligned}$$

Thus,

$$\boxed{E[X^2 + (X+1)^2] = 21}.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q12. Indicator Variables

Ten balls are drawn from an urn containing 17 white balls and 23 black balls. Let X denote the number of white balls drawn. Compute $E[X]$ using indicator variables in two different ways:

(a) using one indicator for each draw,

(b) using one indicator for each white ball.

Solution

(a) One indicator for each draw

Define

$$I_i = \begin{cases} 1, & \text{if the } i\text{-th drawn ball is white,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = I_1 + I_2 + \cdots + I_{10}.$$

For every draw,

$$E[I_i] = P(I_i = 1) = \frac{17}{40}.$$

Therefore,

$$E[X] = 10\left(\frac{17}{40}\right) = \frac{17}{4} = 4.25.$$

(b) One indicator for each white ball

There are 17 white balls. Define

$$J_i = \begin{cases} 1, & \text{if the } i\text{-th white ball is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

Since 10 balls are selected from 40,

$$P(J_i = 1) = \frac{10}{40} = \frac{1}{4}.$$

Hence,

$$E[X] = 17\left(\frac{1}{4}\right) = \frac{17}{4} = 4.25.$$

Thus, both methods give

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\boxed{E[X] = 4.25}.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q13. Geometric Random Variable

A machine produces defective items with probability 0.1. Items are inspected one at a time, independently. Let X be the number of items inspected until the first defective item is found. Find:

(a) $P(X = 5)$,

(b) $P(X \leq 5)$,

(c) $E[X]$.

Solution

X follows a geometric distribution with

$$p = 0.1.$$

(a)

For the first defective item to occur on the fifth inspection, the first four items must be non-defective and the fifth must be defective.

$$P(X = 5) = (0.9)^4(0.1)$$

$$P(X = 5) = 0.06561.$$

Therefore,

$$\boxed{P(X = 5) = 0.06561}.$$

(b)

Using the complement,

$$P(X \leq 5) = 1 - P(X > 5).$$

The event $X > 5$ means that the first five items are all non-defective.
Thus,

$$P(X > 5) = (0.9)^5.$$

Therefore,

$$P(X \leq 5) = 1 - (0.9)^5$$

$$P(X \leq 5) = 0.40951.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Hence,

$$\boxed{P(X \leq 5) = 0.40951}.$$

(c)

For a geometric random variable representing the number of trials until the first success,

$$E[X] = \frac{1}{p}.$$

Therefore,

$$E[X] = \frac{1}{0.1} = 10.$$

Thus,

$$\boxed{E[X] = 10}.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Differentiation Formulas

1. $\frac{d}{dx}(x) = 1$
2. $\frac{d}{dx}(ax) = a$
3. $\frac{d}{dx}(x^n) = nx^{n-1}$
4. $\frac{d}{dx}(\log x) = \frac{1}{x}$
5. $\frac{d}{dx}(e^x) = e^x$
6. $\frac{d}{dx}(a^x) = \log a \ a^x$
7. $\frac{d}{dx}(\sin x) = \cos x$
8. $\frac{d}{dx}(\sin ax) = a\cos ax$
9. $\frac{d}{dx}(\cos x) = -\sin x$
10. $\frac{d}{dx}(\cos ax) = -a\sin ax$
11. $\frac{d}{dx}(\tan x) = \sec^2 x$
12. $\frac{d}{dx}(\tan ax) = a\sec^2 ax$
13. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
14. $\frac{d}{dx}(\cot ax) = -a\operatorname{cosec}^2 ax$
15. $\frac{d}{dx}(\sec x) = \sec x \tan x$
16. $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$
17. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
18. $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
19. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{x^2+1}$
20. $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{x^2+1}$
21. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$
22. $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}$

Integration Formulas

$\int 1\,dx = x$

$\int a\,dx = ax$

$\int x^n\,dx = \frac{x^{n+1}}{n+1}$

$\int \frac{1}{x}dx = \log x$

$\int e^x\,dx = e^x$

$\int a^x\,dx = \frac{a^x}{\log a}$

$\int \cos x\,dx = \sin x$

$\int \cos ax\,dx = \frac{1}{a}\sin ax$

$\int \sin x\,dx = -\cos x$

$\int \sin ax\,dx = -\frac{1}{a}\cos ax$

$\int \sec^2 x\,dx = \tan x$

$\int \sec^2 ax\,dx = \frac{1}{a}\tan ax$

$\int \operatorname{cosec}^2 x\,dx = -\cot x$

$\int \operatorname{cosec}^2 ax\,dx = -\frac{1}{a}\cot ax$

$\int \sec x \tan x\,dx = \sec x$

$\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x$

$\int \frac{1}{\sqrt{1-x^2}}dx = \sin^{-1} x$

$\int \frac{1}{\sqrt{a^2-x^2}}dx = \frac{1}{a}\sin^{-1}\frac{x}{a}$

$\int \frac{1}{x^2+1}dx = \tan^{-1} x$

$\int \frac{1}{x^2+a^2}dx = \frac{1}{a}\tan^{-1}\frac{x}{a}$

$\int \frac{1}{|x|\sqrt{x^2-1}}dx = \sec^{-1} x$

$\int \frac{1}{x\sqrt{x^2-a^2}}dx = \frac{1}{a}\sec^{-1}\frac{x}{a}$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Link for the book: https://minerva.it.manchester.ac.uk/~saralees/statbook3.pdf

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.